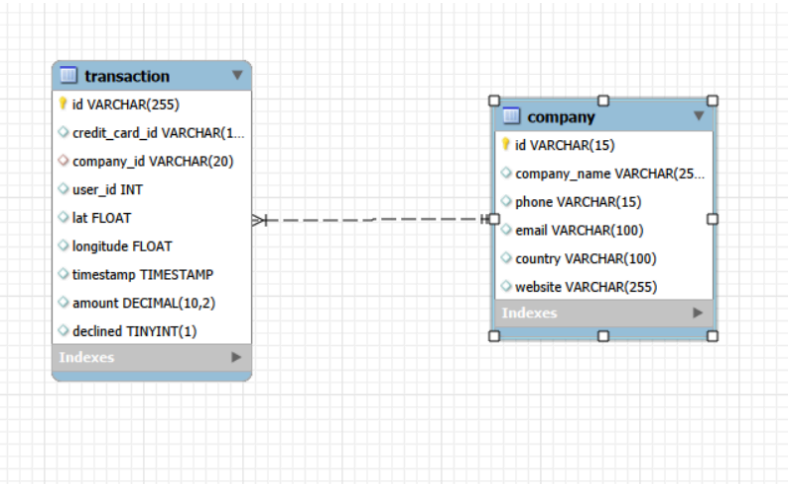
Nivell 1

Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



La base de datos transactions se compone de 2 tablas: transaction y company, que tienen una relación de n a 1, ya que la primera (transacciones) funciona como tabla de hechos registrando transacciones realizadas por distintas empresas que vienen descritas en la tabla de dimensiones company.

Su relación se establece en transaction.company_id y company.id

Tabla transactions

Field	Type	Null	Key
id	varchar(255)	NO	PRI
credit_card_id	varchar(15)	YES	
company_id	varchar(20)	YES	MUL
user_id	int	YES	
lat	float	YES	
longitude	float	YES	
timestamp	timestamp	YES	
amount	decimal(10,2)	YES	
declined	tinyint(1)	YES	

La tabla se describe con las siguientes columnas y tipos de variables:

- id: asigna un identificador único a la transacción, es la primary key (PK), no puede tener valores nulos y es de tipo varchar(255), lo que significa que puede albergar valores de hasta 255 caracteres.
- credit_card_id: contiene variables tipo varchar(15) lo que le permite almacenar hasta 15 caracteres, este campo sí permite valores nulos.

- company_id: este campo funciona como foreign key (FK) en relación con la tabla company, por lo que admite que se repita múltiples veces dentro de él (MUL), es de tipo varchar(20) y admite nulos.
- user_id: es una variable de tipo numérico entero y admite nulos.
- lat: contiene el valor de la latitud y es de tipo numérico float (decimal), admite nulos.
- longitude: al igual que la columna anterior pero almacena la longitud geográfica, también es de tipo float y admite nulos.
- timestamp: almacena en formato timestamp (yyyy-mm-dd hh:mm:ss) los datos de fecha y hora de la transacción. Admite nulos.
- amount: almacena el monto de la transacción, es tipo decimal(10,2) lo que significa que contará con hasta 10 dígitos antes del punto decimal y 2 después de este. Admite nulos.
- declined: de tipo tinyint(1), un valor numérico entero que sólo admite un dígito (0 o 1) por lo que funciona como boleano equivalente a valores True / False.

Tabla company

	Field	Type	Null	Key
▶	id	varchar(15)	NO	PRI
	company_name	varchar(255)	YES	
	phone	varchar(15)	YES	
	email	varchar(100)	YES	
	country	varchar(100)	YES	
	website	varchar(255)	YES	

La tabla se describe con las siguientes columnas y tipos de variables:

- id: es la primary key (PK) de la tabla, asigna un identificador a cada empresa, es de tipo varchar(15) por lo que admite hasta 15 caracteres, no admite nulos.
- company_name: almacena el nombre de la empresa, es de tipo varchar(255) admite hasta 255 caracteres, admite nulos.
- phone: contiene el número de teléfono de la empresa, es de tipo varchar(15), para hasta 15 caracteres, admite nulos.
- email: almacena el email de la empresa, es de tipo varchar(100), para hasta 100 caracteres, admite nulos.
- country: contiene el país de la empresa, es de tipo varchar(100) al igual que la columna anterior, admite nulos.
- website: almacena la dirección web de la empresa, es de tipo varchar(255) para hasta 255 caracteres, admite nulos.

Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

Llistat dels països que estan generant vendes.

```
SELECT DISTINCT c.country
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id;
```

country
Germany
Australia
United States
New Zealand
Norway
United Kingdom
Italy
Belgium
Sweden
Ireland
China
Canada
France
Netherlands
Spain

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Result 14 — Read Only

Output — Action Output

#	Time	Action	Message	Duration / Fetch
4	23:35:56	SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id LIMIT 0, 2000	15 row(s) returned	0.078 sec / 0.000 sec

Des de quants països es generen les vendes.

```
SELECT COUNT(DISTINCT c.country)
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id;
```

COUNT(DISTINCT c.country)
15

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Result 14 — Read Only

Output — Action Output

#	Time	Action	Message	Duration / Fetch
5	23:35:57	SELECT COUNT(DISTINCT c.country) FROM company c JOIN transaction t ON c.id = t.company_id LIMIT 0, 2000	1 row(s) returned	0.188 sec / 0.000 sec

Identifica la companyia amb la mitjana més gran de vendes.

```
SELECT c.company_name, ROUND(AVG(amount),2) AS media_ventas
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id
 GROUP BY c.id
 ORDER BY media_ventas DESC
 LIMIT 1;
```

company_name	media_ventas
Ac Fermentum Incorporated	284.87

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Result 14 — Read Only

Output — Action Output

#	Time	Action	Message	Duration / Fetch
6	23:35:57	SELECT c.company_name, ROUND(AVG(amount),2) AS media_ventas FROM company c JOIN transaction t ON c.id = t.company_id GR...	1 row(s) returned	0.641 sec / 0.000 sec

Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
SELECT t.id
FROM transaction t
WHERE company_id IN (SELECT id
                     FROM company
                     WHERE country = 'Germany');
```

id
00138D3B-206D-4C03-94B7-63A2676EB9B4
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9
00201A11-2E62-44C4-941D-198FC8DB77F0
00235618-0A5C-4D49-9DCB-B3A9405D8923
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38
00687139-48B2-4FFA-8E73-B20376F04AB4
0074F4DD-32F1-4827-8758-55896314623A
00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9
00BE09D4-6920-47D8-ABE8-325E2269829D
00DA0383-E048-4577-8ED1-3C56C258FF2F

Output

#	Time	Action	Message	Duration / Fetch
7	23:35:58	SELECT t.id FROM transaction t WHERE company_id IN (SELECT id FROM company WHERE country = 'Germany') LIMIT 0, 20...	2000 row(s) returned	0.000 sec / 0.000 sec

Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
SELECT id,
       company_name
  FROM company
 WHERE id IN (SELECT company_id
              FROM transaction
             WHERE amount > (
                   SELECT AVG(amount) AS media_total_transactions
                     FROM transaction
                   )
             );
```

id	company_name
b-2222	Ac Fermentum Incorporated
b-2226	Magna A Neque Industries
b-2230	Fusce Corp.
b-2234	Convallis In Incorporated
b-2238	Ante Iaculis Nec Foundation
b-2242	Donec Ltd
b-2246	Sed Nunc Ltd
b-2250	Amet Nulla Donec Corporation
b-2254	Nascetur Ridiculus Mus Inc.
b-2258	Vestibulum Lorem PC
b-2262	Gravida Sagittis LLP

Output

#	Time	Action	Message	Duration / Fetch
8	23:35:58	SELECT id, company_name FROM company WHERE id IN (SELECT company_id FROM transaction WHERE amount > (SELECT AVG(a...	100 row(s) returned	0.125 sec / 0.000 sec

Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
SELECT id
  FROM company
 WHERE id NOT IN (
                  SELECT DISTINCT company_id
                    FROM transaction
                  );
```

id
NULL

Output

#	Time	Action	Message	Duration / Fetch
9	23:35:59	SELECT id FROM company WHERE id NOT IN (SELECT DISTINCT company_id FROM transaction) LIMIT 0, 2000	0 row(s) returned	0.063 sec / 0.000 sec

Nivell 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
SELECT DATE(timestamp) AS day,
       SUM(amount) AS total_amount
  FROM transaction
 GROUP BY DATE(timestamp)
 ORDER BY total_amount DESC
 LIMIT 5;
```

day	total_amount
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

#	Time	Action	Message	Duration / Fetch
10	23:35:59	SELECT DATE(timestamp) AS day, SUM(amount) AS total_amount FROM transaction GROUP BY DATE(timestamp) ORDER BY total...	5 row(s) returned	0.140 sec / 0.000 sec

Exercici 2

Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
SELECT c.country,
       ROUND(AVG(num_trans.total_transactions),2) AS average_transactions
  FROM company c
  JOIN (
       SELECT company_id,
              COUNT(id) AS total_transactions
         FROM transaction
        GROUP BY company_id
      ) AS num_trans
    ON c.id = num_trans.company_id
 GROUP BY c.country
 ORDER BY average_transactions DESC;
```

country	average_transactions
Germany	1661.38
France	1566.67
United Kingdom	1530.67
Italy	1518.67
Spain	1515.00
Netherlands	1505.50
Sweden	1501.18
China	458.00
United States	448.44
Ireland	443.17

#	Time	Action	Message	Duration / Fetch
11	23:35:59	SELECT c.country, ROUND(AVG(num_trans.total_transactions),2) AS average_transactions FROM company c JOIN (SELECT com...	15 row(s) returned	0.047 sec / 0.000 sec

Exercici 3

En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute". Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

Mostra el llistat aplicant JOIN i subconsultes.

```
SELECT t.id AS transaction_id_NI_competitors,
       c.company_name AS NI_competitors
  FROM transaction t
  JOIN company c
    ON t.company_id = c.id
 WHERE c.company_name <> "Non Institute"
       AND country = (
                      SELECT country
                        FROM company
                       WHERE company_name = "Non Institute"
                      );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

transaction_id_NI_competitors	NI_competitors
008629B4-C9A9-406C-A3D2-71FDA47BC546	Sed Nunc Ltd
00B72BA4-54A3-4B8E-B13F-2D57535AA17A	Sed Nunc Ltd
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	Sed Nunc Ltd
023FFCE8-E618-4938-BF56-C8DF80540ADD	Sed Nunc Ltd
02683BEB-EF91-4564-957B-D6F1662AB7C5	Sed Nunc Ltd
02C2F29E-CEF2-4C1E-A594-F476E8F279C0	Sed Nunc Ltd
02F468DC-426C-47C2-8B0A-D8B25B7A81AF	Sed Nunc Ltd
0306BE3B-817B-4A49-934E-0E439291A104	Sed Nunc Ltd
0347BFE6-8EB5-4387-B187-0E78E8F2B8FB	Sed Nunc Ltd
03AFBD0F-DC97-4BD3-9C57-6A6DB78026FD	Sed Nunc Ltd

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	23:35:59	SELECT t.id AS transaction_id_NI_competitors, c.company_name AS NI_competitors FROM transaction t JOIN company c ON t.co...	2000 row(s) returned	0.000 sec / 0.000 sec

Mostra el llistat aplicant solament subconsultes.

```
SELECT t.id AS transaction_id_NI_competitors
  FROM transaction t
 WHERE company_id IN (
                      SELECT id
                        FROM company
                       WHERE country = (
                                        SELECT country
                                          FROM company
                                         WHERE company_name = "Non Institute"
                                        )
                             AND company_name <> "Non Institute"
                      );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

transaction_id_NI_competitors
008629B4-C9A9-406C-A3D2-71FDA47BC546
00B72BA4-54A3-4B8E-B13F-2D57535AA17A
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C
023FFCE8-E618-4938-BF56-C8DF80540ADD
02683BEB-EF91-4564-957B-D6F1662AB7C5
02C2F29E-CEF2-4C1E-A594-F476E8F279C0
02F468DC-426C-47C2-8B0A-D8B25B7A81AF
0306BE3B-817B-4A49-934E-0E439291A104
0347BFE6-8EB5-4387-B187-0E78E8F2B8FB
03AFBD0F-DC97-4BD3-9C57-6A6DB78026FD

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	23:35:59	SELECT t.id AS transaction_id_NI_competitors FROM transaction t WHERE company_id IN (SELECT id FROM company WHERE co...	2000 row(s) returned	0.000 sec / 0.000 sec

Nivell 3

Exercici 1

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
SELECT c.company_name,
       c.phone,
       DATE(t.timestamp) AS date,
       t.amount
  FROM company c
  JOIN transaction t
    ON t.company_id = c.id
 WHERE amount BETWEEN 350 AND 400
       AND DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
 ORDER BY amount DESC;
```

company_name	phone	date	amount
Aliquam PC	01 45 73 52 16	2024-03-13	399.84
Auctor Mauris Vel LLP	08 09 28 74 14	2018-07-20	399.51
At Pede Corp.	06 14 48 33 15	2015-04-29	390.69
Aliquam PC	01 45 73 52 16	2024-03-13	388.29
Orci Adipiscing Limited	03 18 00 77 81	2018-07-20	373.71
Fringilla LLC	08 29 15 93 57	2015-04-29	367.62
Pede Cum Ltd	07 62 26 48 38	2018-07-20	356.87
Auctor Mauris Vel LLP	08 09 28 74 14	2024-03-13	353.75

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Result 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	23:35:59	SELECT c.company_name, c.phone, DATE(t.timestamp) AS date, t.amount FROM company c JOIN transaction t ON t.company_id = c.id W...	8 row(s) returned	0.078 sec / 0.000 sec

Exercici 2

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen més de 400 transaccions o menys.

```
SELECT company_id,
       COUNT(id) AS total_transactions,
  CASE
       WHEN COUNT(id) >= 400 THEN True
       WHEN COUNT(id) < 400 THEN False
   END AS 'More than 400'
  FROM transaction
 GROUP BY company_id
 ORDER BY COUNT(id);
```

company_id	total_transactions	More than 400
b-2362	380	0
b-2286	397	0
b-2470	399	0
b-2578	402	1
b-2446	407	1
b-2226	410	1
b-2526	411	1
b-2298	418	1

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | transaction 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | transaction 12 | Result 13 | Result 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	23:35:59	SELECT c.company_name, c.phone, DATE(t.timestamp) AS date, t.amount FROM company c JOIN transaction t ON t.company_id = c.id W...	8 row(s) returned	0.078 sec / 0.000 sec
15	23:35:59	SELECT company_id, COUNT(id) AS total_transactions, CASE WHEN COUNT(id) >= 400 THEN True WHEN COUNT(id) < 400 THEN False E...	100 row(s) returned	0.063 sec / 0.000 sec